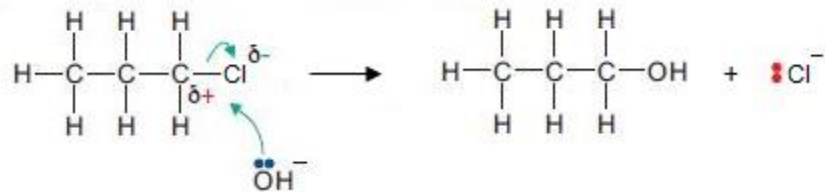


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12.	(a)	(i)		add aqueous NaOH and then (neutralise with) HNO ₃ (1) add aqueous AgNO ₃ (1) yellow precipitate for iodine, cream for bromine and white for chlorine (1)	3			3		3
		(ii)		Y is C ₄ H ₉ Cl / chlorobutane (1) award (1) for any explanation supported by a simple calculation e.g. showing that chlorine accounts for 38% of mass of chlorobutane showing that bromine (and iodine) account for greater than 40% of mass of their respective halogenoalkanes showing that the hydrocarbon chain has a mass of 57 and only chlorine has an A _r value less than 40% of the total mass			2	2		
	(b)	(i)		 diagram showing δ ⁺ and δ ⁻ (1) lone pair on OH ⁻ (1) curly arrows from lone pair and to break C—Cl bond (1) propan-1-ol and Cl ⁻ formed (1)	4			4		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		nucleophilic substitution / hydrolysis	1			1		
	(c)	(i)		reflux/heat with ethanolic / alcoholic sodium hydroxide	1			1		
		(ii)		$\text{CH}_2=\text{CHCH}_2\text{CH}_2\text{CH}_3$ (1) $\text{CH}_3\text{CH}=\text{CHCH}_2\text{CH}_3$ (1) only credit one <i>E/Z</i> isomer award (1) for both correct names if no structures given pent-1- and pent-2-ene		2		2		
				Question 12 total	9	2	2	13	0	3